

Eugene Lafayette Buchanan

Quality Assurance Analyst — Systems Applications Integration and Development (SCDHHS 13582)

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Willing to relocate to Columbia, SC prior to start at own expense for fully onsite role (5 days/week, 1801 Main Street)

Professional Summary

Quality Assurance Analyst with 25+ years in Quality Assurance (QA) and Quality Control (QC), including Senior QA Engineer, Test Automation Lead, and Product Owner roles. I translate business requirements and policy documentation into test cases and scenarios, build them into test systems, and execute Smoke, System Integration, and End-to-End testing with regression, negative testing, and usability coverage. Selenium Automation Framework practitioner (Cypress, Selenium, Postman, JUnit, Qase), with nightly build automation, CI/CD quality pipelines, health dashboards, defect tracking in Jira and Qase, and status reporting to developers and leadership. Experienced in large enterprise and e-business systems, document-heavy workflows, software evaluation and migration including medical-records software, and state certification submissions. Strong written and oral communication in English; requirements, Use Cases, test plans, test scripts and scenarios, test reports, walkthroughs, and inspections. Seeking SCDHHS Systems Applications Integration and Development QA role, Columbia, SC.

Core Skills — Mapped to SCDHHS Requirements

QA and QC (5+ years, 25+ total): test plans; QA and User Acceptance Testing (UAT) processes; test cases and scenarios from business requirements and policy documentation; Smoke test, System Integration Test, End-to-End test; regression testing; negative testing; usability testing; reliability and stability testing; security-flaw and defect assessments; performance and load testing; accessibility and localization testing

Test automation (Selenium side of Sahi Pro/Selenium requirement): Selenium Automation Framework (satisfies Sahi Pro/Selenium requirement via Selenium side); design and develop test automation scripts per test automation guidelines; Cypress; JUnit; Postman; nightly build test script automation; CI/CD pipeline quality gates (Apache, REST, SQL, AWS, React); health dashboards and alert systems; Qase and Jira for complementary test management and defect tracking

Formal test design: Equivalence Class Partitioning; Pairwise Analysis; orthogonal-array test design (PICT-equivalent combinatorial approach); requirements-to-test traceability

Process and governance: structured test governance (Product Development Review [PDR] life cycles, backlog with KPI tracking, Standard Operating Procedure [SOP] guides, SOC 2 compliance research and implementation); exposure to maturity-model terminology (Systems and software Process Improvement and Capability Evaluation [SPICE], Test Process Improvement [TPI], Test Maturity Model integration [TMMi], Capability Maturity Model Integration [CMMI] Organizational Process Focus [OPF], Organizational Process Definition [OPD], Process and Product Quality Assurance [PPQA])

Defect and test management: Jira; Confluence; Qase; defect database maintenance (defects, reviews, functional improvements); test reports; progress-against-target reporting and exception resolution; vendor test plan and test result review; test plan, test script, and scenario review and guidance

Methodologies: Agile delivery and standard Software Development Life Cycle (SDLC) waterfall; cross-functional collaboration with architects, technicians, developers, and management

Integrations, workflows, documents: REST APIs, webhooks, SFTP, SAML, OAuth, SCORM, LTI; document-management-adjacent work (medical-records software evaluation and migration, learning content workflows, state attendance Extract Transform Load [ETL] with certification); enterprise integrations (SAP, Oracle, Workday, Coursera, UKG, Zoom)

Languages and stack: JavaScript, TypeScript, Python, Rust, PHP, Node.js, React, SQL; AWS (Elastic Compute Cloud [ECS pattern], Simple Storage Service [S3]); Docker; Git; Bitbucket; Elasticsearch; Scorm Cloud

Communication: superb written and oral English; requirements and Use Cases; stakeholder feedback documentation; testing-outcome reports to management; plain-language status updates

Work Experience

Senior QA Engineer / Test Automation Lead / Product Owner — Integrations

Knowledgecity LLC, Remote, Dec 2020 – Aug 2025

- Translated business requirements and integration specs into test cases and scenarios; built them into Cypress/Selenium/JUnit/Postman suites tracked in Qase, Jira, and Confluence.
- Developed test plans and QA and User Acceptance Testing (UAT) processes for product management and software development teams; provided guidance and review of test plans, test scripts and scenarios, and test reports.
- Designed and supported user interface testing and end-to-end regression testing software with health dashboards and alert systems for complex deployment scenarios; tested for reliability and stability across integration, frontend, backend, mobile, security, performance, load, API, database, functionality, usability, accessibility, and localization.
- Ran Smoke, System Integration, and End-to-End coverage including regression, negative testing, and usability; proactively identified issues and action items, reported impediments, and coordinated with leadership and technical resources.
- Maintained the defect database in Qase and Jira; published status reports on open issues and gaps found during the test phase and articulated details to development teams; tracked progress against targets and resolved exceptions.
- Performed reviews, walkthroughs, and inspections under established governance (PDR life cycles, backlog with KPIs, SOPs, SOC 2 controls); reviewed vendor-style integration test plans and results for SAP, Oracle, Workday, Coursera, UKG, and Zoom (SAML, OAuth, SFTP, REST APIs, webhooks, SCORM, LTI).
- Collaborated with architects, technicians, and management; documented stakeholder feedback and testing outcomes for management; recommended solutions to maximize performance and efficiency.
- Helped develop an AI support chatbot with retrieval-augmented generation (RAG) context retrieval and guardrails; applied Equivalence Class Partitioning and Pairwise/orthogonal combinations when designing regression and integration scenario matrices.

Business Consultant — Campus IT, Software Evaluation and Migration

Medico Investments LLC / Foremost Senior Care / Foremost Organization, Hesperia, CA, 2013 – 2016

- Evaluated and migrated accounting and medical-records software — document-heavy workflow experience directly transferable to enrollment, eligibility, and claims-adjacent document management and workflows (no Medicaid eligibility-determination claim).
- Provided campus IT support across servers and PCs: wrote security and backup policies, upgraded video surveillance, installed campus-wide wireless; set up sound/video and ran community events.
- Documented requirements and Use Cases in plain English; coordinated nonprofit filings across federal, state, and city levels, modeling policy-document-to-execution traceability.

Computer Service Technician — Geek Squad (Retail Service Desk)

Best Buy, Victorville, CA, 2007

- Ran the retail service desk: logged tickets, triaged walk-ins, tracked units through check-in, repair, and pickup; gave plain-language status updates — frontline equivalent of ticket lifecycle management.
- Set up new PCs, migrated user data, resolved virus, operating system, and hardware issues; handled customer service, payments, and refunds.

Freelance IT Consultant

Amp Computer, Apple Valley, CA, 2003 – 2011

- Sold and supported servers and PCs; configured Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) systems, office automation, and custom solutions for small-business e-business needs.

IT Manager

Lucerne Valley Unified School District, Lucerne Valley, CA, 2002

- Supported four campuses (Wide Area Network [WAN]/Local Area Network [LAN], NetWare, Windows, Linux, virtualized desktops, audio/video).
- Met state attendance submission requirements via data Extract, Transform, and Load (ETL) with table locking and certification — state-reporting compliance and workflow discipline transferable to eligibility-system testing.

Senior Software Engineer II (SDET Practice)

RealNetworks, Seattle, WA, 1999 – 2001

- Software Development Engineer in Test (SDET) quality assurance across Linux, Unix, Windows, and embedded systems for streaming media platform, servers, and mobile/consumer appliances including TFRCP patent work.
- Large-enterprise and e-business systems experience; global presales engineering with travel to Japan, Korea, and the continental United States; trained new hires.

Earlier Software Test Engineering and Support Roles

Microsoft, IBM, Keene Inc., 1996 – 1999

- Microsoft, Redmond, WA (1999): Senior Software Test Engineer IV, Windows Media Server versions 4 and 5; nightly build test script automation per test automation guidelines.
- IBM, Kirkland, WA (1998): Senior Test Engineer III for high-availability servers; Working knowledge of certification-style governance via WHQL certification for cluster failover.
- Microsoft, Redmond, WA (1997): Software Test Engineer II for Windows 98 and Windows NT; original equipment manufacturer (OEM) setup and hardware/driver verification.
- Keene Inc., Seattle, WA (1996): Technical Support Agent II for Windows 95 and Windows NT; frontline and escalated support ticketing with defect-style queue discipline.

Open-Source Contributor — Test-Relevant Engineering Rigor

ZeroClaw Labs, github.com/zeroclaw-labs/zeroclaw, 32.8k GitHub stars, 2026 – Present

- PR #7946 (merged Jul 2026): context-window meter across TUI, gateway, and CLI covering 9 providers; demonstrates requirements-to-implementation traceability and reviewer-driven quality gates.
- PR #8966 (open): live provider identity on usage events for attribution across fallback — systematic edge-case and failure-path testing mindset.
- PR #8337 (closed, superseded by #10269): agent lifecycle observer over JSON-RPC; cleared four rounds of maintainer review; requirements adopted into vendor-neutral design.

Education

- Associate Degree, Victor Valley College — qualifies under requisition clause "Bachelor's degree in a technical, business, or healthcare field, or 5+ years of equivalent experience in Quality Assurance" via 25+ years QA/QC practice.
- California Notary Commission
- Toastmasters International (written and oral communication)

Availability and Work Location

Available for fully onsite work at 1801 Main Street, Columbia, SC 29201, 5 days/week. Not currently a South Carolina resident; willing to relocate to South Carolina prior to starting the role at own expense. Very strong proficiency in English.